

Distribuciones Continuas

Nombre	Notación	Soporte	Parametros	FDP	FMP	$E(X)$	$VAR(X)$	$M_X(t)$
Exponencial	$X \sim \text{Exp}(\lambda)$	$0 < x$	$0 < \lambda$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$
Normal	$X \sim N(\mu, \sigma^2)$	$x \in \mathbb{R}$	$-\infty < \mu < +\infty$ $0 < \sigma^2 < +\infty$	$\frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma^2}}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	μ	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Uniforme	$X \sim \mathcal{U}_{a,b}$	$x \in [a, b]$	$-\infty < a < +\infty$ $-\infty < b < +\infty$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{1}{2}(a+b)$	$\frac{1}{12}(b-a)^2$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$
Gamma	$X \sim \text{Gamma}(a, \lambda)$	$x \in [0, \infty)$	$0 < \alpha$ $0 < \lambda$	$\frac{\lambda^a}{\Gamma(a)} x^{\alpha-1} e^{-\lambda x}$	$\frac{\lambda^a}{\Gamma(a)} \gamma(a, \lambda x)$	$\frac{a}{\lambda}$	$\frac{a}{\lambda^2}$	$(1 - \frac{t}{\lambda})^{-a}, \quad t < \lambda$
Beta	$X \sim \text{Beta}(\alpha, \beta)$	$x \in [0, 1]$	$0 < \alpha$ $0 < \beta$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\text{B}(\alpha, \beta)}$	...	$\frac{\alpha}{\alpha+\beta}$		
Pareto	$X \sim \text{Pareto}(\alpha, \beta)$	$a < x < \infty$	$0 < \alpha$ $0 < \beta$	$\frac{\beta \alpha^\beta}{x^{\beta+1}}$	$1 - (\frac{\alpha}{x})^\beta$	$\frac{\alpha \beta}{\beta-1}$	$\frac{\alpha^2 \beta}{(\beta-1)^2(\beta-2)}$	No existe
Cauchy	$X \sim \text{Cauchy}(\theta, \sigma)$	$x \in \mathbb{R}$	$-\infty < \theta < +\infty$ $0 < \sigma$	$\frac{1}{\pi \sigma [1 + (\frac{x-\theta}{\sigma})^2]}$	$\frac{1}{\pi} \arctan(\frac{x-\theta}{\sigma}) + \frac{1}{2}$	No existe	No existe	No existe
Chi cuadrado	$X \sim X^2_{(k)}$	$0 < x < \infty$	$k \in \mathbf{N}$	$\frac{1}{2^{k/2} \Gamma(k/2)} x^{k/2-1} e^{-x/2}$				

Nombre	Fisher	priori conjugada (bayes)	Posteriori	Forma Exponencial
Exponencial	$\frac{1}{\lambda^2}$			
Normal	$\begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}$	$\lambda \sim \text{Gamma}(a, b)$		$\frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{\mu^2}{2\sigma^2}} e^{\frac{\mu}{\sigma^2}x - \frac{1}{2\sigma^2}x}$
Uniforme	no existe			
Gamma	$\begin{pmatrix} \psi^{(1)}(\alpha) & -\lambda^{-1} \\ -\lambda^{-1} & \alpha\lambda^{-2} \end{pmatrix}$			
Beta	muy feo			
Pareto	$\mathcal{I}(\alpha, \beta) = \begin{bmatrix} \frac{\beta^2}{\alpha^2} & 0 \\ 0 & \frac{1}{\beta^2} \end{bmatrix}$			
Cauchy	$\frac{1}{2\sigma^2}$			

Distribuciones Discretas

Nombre	Notación	Soporte	Parametros	FDP	FMP	μ	VAR(X) σ^2	$M_X(t)$
Bernoulli	$X \sim Bernoulli(p)$	$x = 0, 1$	$p \in [0, 1]$	$p^x(1-p)^{1-x}$	$0 \quad x < 0$ $1-p \quad 0 \leq x < 1$ $1 \quad 1 \leq x$	p	p	$(1-p) + pe^t$
Binomial	$X \sim Binomial(n, p)$	$x \in \{0, \dots, n\}$	$p \in [0, 1]$	$\binom{n}{x} p^x (1-p)^{n-x}$	...	np	$np(1-p)$	$(1-p + pe^t)^n$
Uniforme	$X \sim \mathbf{U}_{\{a,b\}}$	$x \in \{a, a+1, \dots, b-1, b\}$	$a, b \in \mathbf{N}_0$ $n = b - a + 1$	$\frac{1}{n}$	$\frac{\lfloor x \rfloor - a + 1}{n}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2 - 1}{12}$	$\frac{e^{at} - e^{(b+1)t}}{n(1-e^t)}$
Geometrica	$X \sim Geometric(p)$	$x = 1, 2, 3, \dots$	$0 \in [0, 1]$	$p(p-1)^{x-1}$	$1 - (1-p)^{\lfloor x \rfloor}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Hipergeometrica	$Hypergeoemtric(N, M, K)$							
Poisson	$X \sim Pois(\lambda)$ o $P(\lambda)$	$x \in [0, 1, 2, \dots]$	$\lambda \in (0, \infty)$	$\frac{e^{-\lambda} \lambda^x}{x!}$	...	λ	λ	$e^{\lambda(e^1-1)}$

Nombre	Fisher	priori conjugada (bayes)	Posteriori	Forma Exponencial
Bernoulli	$\frac{1}{p(p-1)}$			
Binomial	$\frac{n}{p(1-p)}$			
Uniforme	no existe			
Geometrica	$\frac{1}{p^2(1-p)}$			
Hipergeometrica	?			
Poisson	$\frac{1}{\lambda}$			